From hidden paths to belief geometry

HMM/MSP synthesis · after Notebook 1 CORE 4 · 12–15 minutes plus questions

ILIAD Intensive · Computational Mechanics

What did you build?

HMM edges → $\Pr(w)$ → $\eta^{(w)}$ → F_x → **belief geometry** → future predictions

Linear propagation

Matrix products sum hidden paths.

Nonlinear step

Normalization depends on the current belief.

Change of view

Geometry is the same posteriors seen as points and maps.

Two machines answer two questions

Hidden Markov model

Question: how can a latent mechanism generate the observable stream?

State: hidden phase S_t.

Dynamics: symbol-labelled stochastic edges.

Mixed-state presentation

Question: how does an observer update uncertainty from the symbols?

State: reachable posterior $\eta^{(w)}$.

Dynamics: normalized maps F_x.

The MSP is an inference process induced by the generator—not another hidden physical machine.

One normalization creates the observer dynamics

$$\underbrace{\pi T^{(w)}}_{\text{unnormalized forward row}} \xrightarrow{\text{sum}} \Pr(w) = \pi T^{(w)} \mathbf{1} \xrightarrow{\text{normalize}} \eta^{(w)} = \frac{\pi T^{(w)}}{\pi T^{(w)} \mathbf{1}}.$$

After a new symbol x arrives, reuse the posterior as the next prior:

$$F_x(\eta) = \frac{\eta T^{(x)}}{\eta T^{(x)} \mathbf{1}}.$$

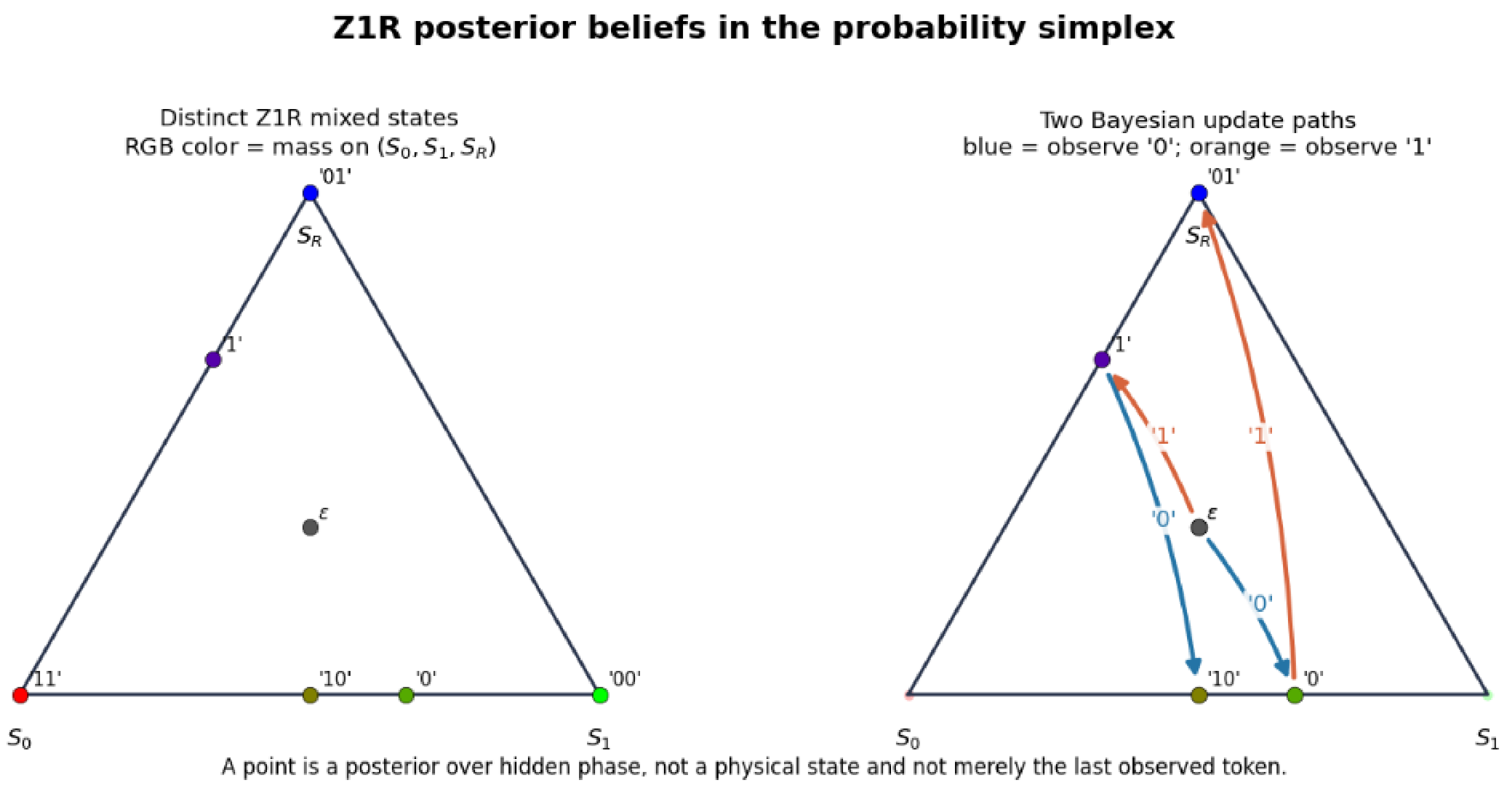
Before normalization

Matrix propagation is linear.

After normalization

The denominator bends straight mixtures into nonlinear dynamics.

Bayes becomes motion inside a simplex



Each point is a posterior over hidden phase. Each arrow is “observe one more symbol, then renormalize.”

Stop: do the Mess3 calculation

DO NOT REVEAL THE RESULT YET

Return to the Mess3 synthesis in Notebook 1. Normalize the visible rows for each symbol and calculate at least one two-map composition—or complete that calculation together.

Where does each one-symbol Bayesian update send the hidden-state simplex?

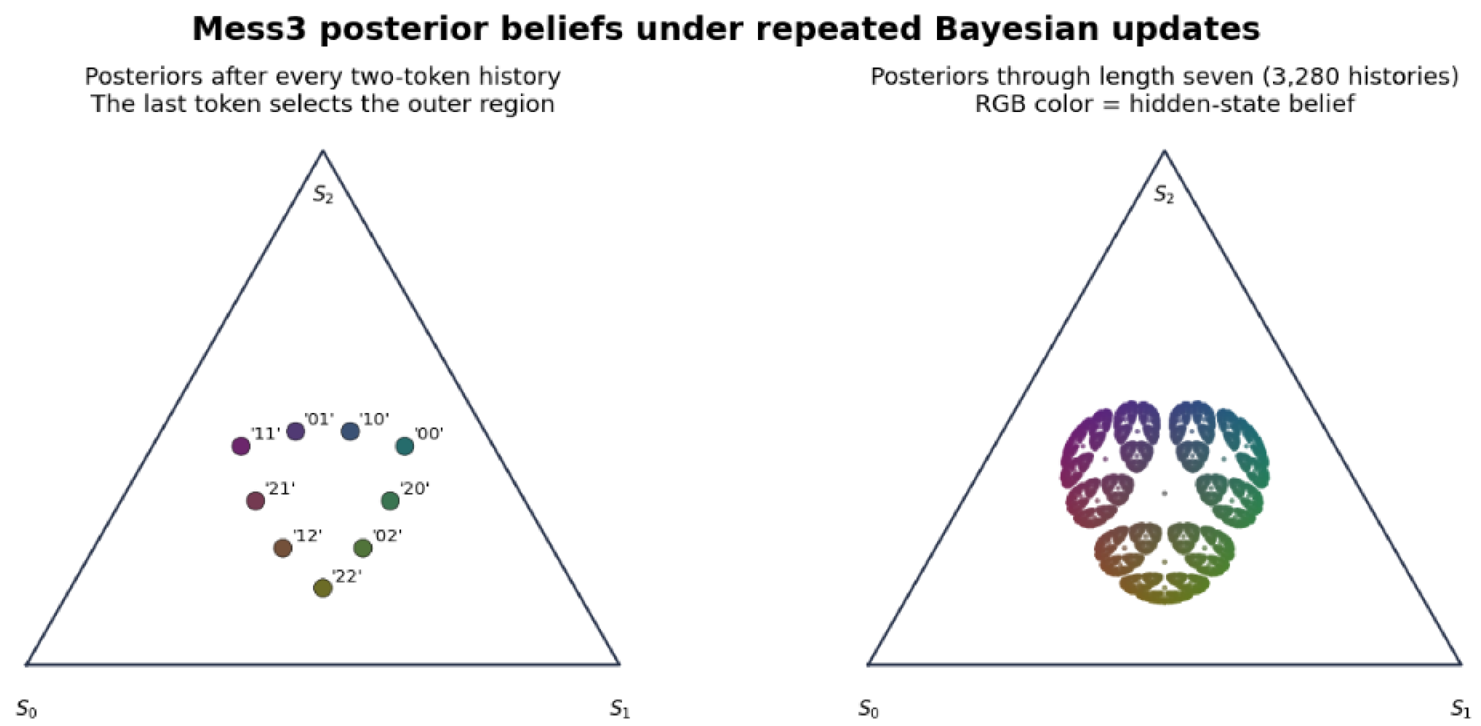
One symbol selects an image of the simplex

observe 0 → $F_0(\Delta^2)$

observe 1 → $F_1(\Delta^2)$

observe 2 → $F_2(\Delta^2)$

Repeated Bayes maps carve predictive geometry



Every dot is a posterior belief. Three hidden generator states do not mean three predictive states.

Three hidden generator states define the ambient triangle—not the number of reachable observer beliefs.

Same next token, different future

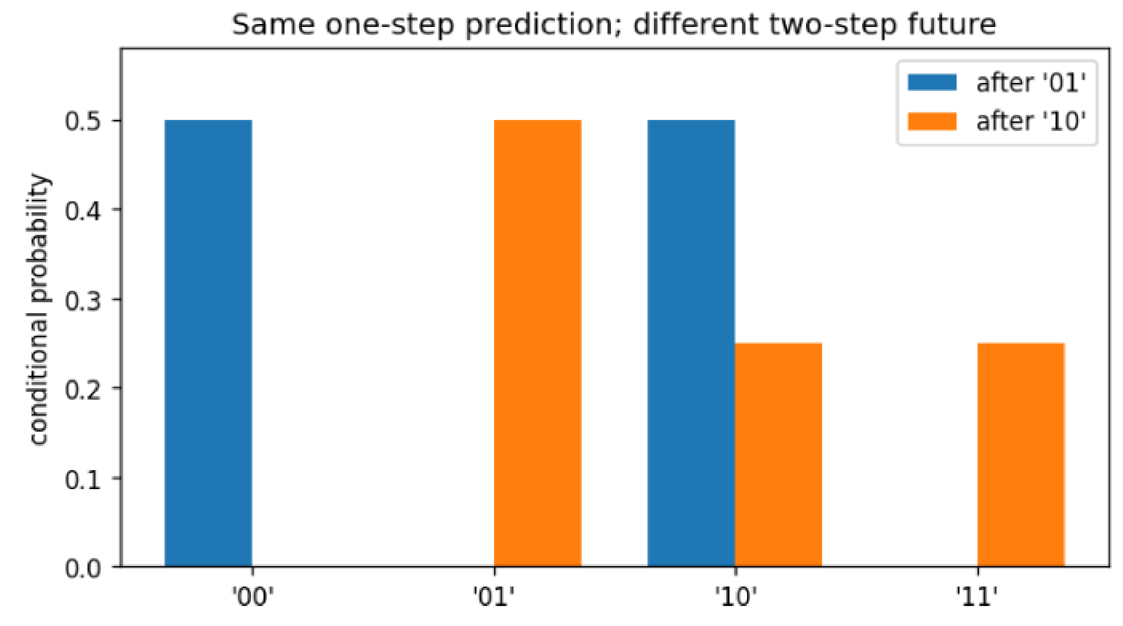
$$\eta^{(01)} = (0, 0, 1), \qquad \eta^{(10)} = (1/2, 1/2, 0).$$

Both give

$$\Pr\left(X_{t+1} = 0, 1 \mid h\right) = (1/2, 1/2),$$

but disagree over two-token futures.

> **The current next-token vector is not generally sufficient as the *sole recursively retained state*.**



What follows—and what does not

Established

A chosen HMM posterior is sufficient to predict every future word generated by that realization.

Qualified

Belief coordinates are model-relative and may contain redundant distinctions.

Not established

A full-prefix transformer need not persistently store this posterior; it can recompute or use other predictive coordinates.

AFTERNOON PUZZLE

If next-token training rewards future-sensitive computation, what trace—if any—should that leave in transformer activations?